Virtual Network Simulation: Design, Monitoring and Troubleshooting

Building, monitoring and troubleshooting a routed network in Cisco Packet Tracer

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Programme	Yuva virtual internship - IT / Network Support task
Simulation software	Cisco Packet Tracer
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Executive summary

This report documents a small-scale virtual network built in Cisco Packet Tracer to practise establishing, maintaining and troubleshooting network systems. The network joins two LANs through two routers running RIPv2. Monitoring was implemented with ping, traceroute, Syslog and SNMP, and packet flow was inspected in Simulation mode. Three realistic faults (a shut-down router interface, a wrong default gateway and a removed routing process) were introduced, diagnosed and resolved, and the evidence for each is shown in the sections below.

- **Topology:** 2 routers, 2 switches, 4 PCs and 1 server across two subnets joined by a /30 link.
- **Monitoring:** ICMP (ping), traceroute, Syslog to Server0, SNMPv2c queries from PC1, and Simulation-mode packet capture.
- **Troubleshooting:** 3 incidents documented with symptom, diagnosis, root cause, resolution and verification.

1. Introduction and objectives

Network maintenance depends on three habits: knowing what a healthy network looks like, watching it continuously, and following a repeatable method when something breaks. The objective of this task was to build a small environment where all three can be practised safely.

- Select suitable simulation software and justify the choice.
- Design a topology with routers, switches and at least two end nodes.
- Implement monitoring tools (ping, traceroute, SNMP, Syslog) inside the simulation.
- Simulate network issues, then document the troubleshooting steps and resolutions.
- Compile the configuration, evidence and analysis into this report.

2. Simulation software selection

Three common options were compared before starting.

Criterion	Cisco Packet Tracer	GNS3	Virtual machines (VirtualBox / VMware)
Cost	Free with a Networking Academy account	Free, but needs router images	Free hypervisor, OS licences vary
Hardware needs	Low (runs on a normal laptop)	Medium to high	High (one VM per device)
Learning curve	Gentle; graphical, built-in devices	Steeper; image and server setup	Steep; manual OS and network setup
Router / switch CLI	Cisco IOS-style CLI	Real IOS images	Depends on software router
SNMP, Syslog, Simulation mode	Built in (MIB Browser, Syslog service, packet capture)	External tools needed	External tools needed

Decision: Cisco Packet Tracer was selected. It provides everything the task requires in a single tool - IOS-style configuration, an SNMP MIB Browser, a Syslog service on the server and a packet-level Simulation mode - and it runs comfortably on a standard laptop. Its main limitation is that devices are simulated, so a few advanced IOS features are unavailable; none of them were needed here.

3. Network design

3.1 Topology

The design uses two LANs, each with its own switch and default gateway, connected by a point-to-point /30 link between two routers. A server in LAN1 acts as the Syslog collector, and PC1 acts as the SNMP management station. Keeping the monitoring tools in LAN1 means traffic from Router2 must cross the router link to reach them, which also tests routing.

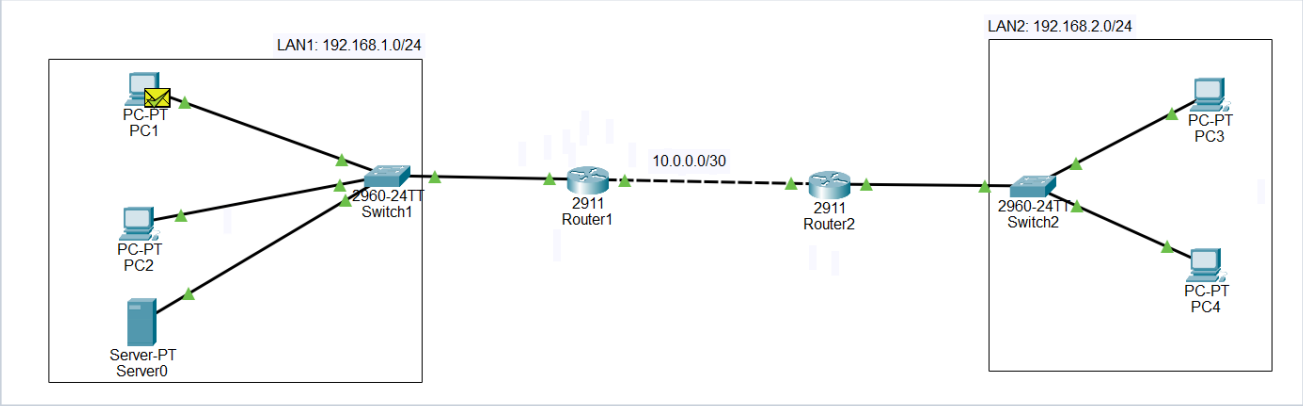


Figure 1: Network topology as built in Cisco Packet Tracer (Logical view).

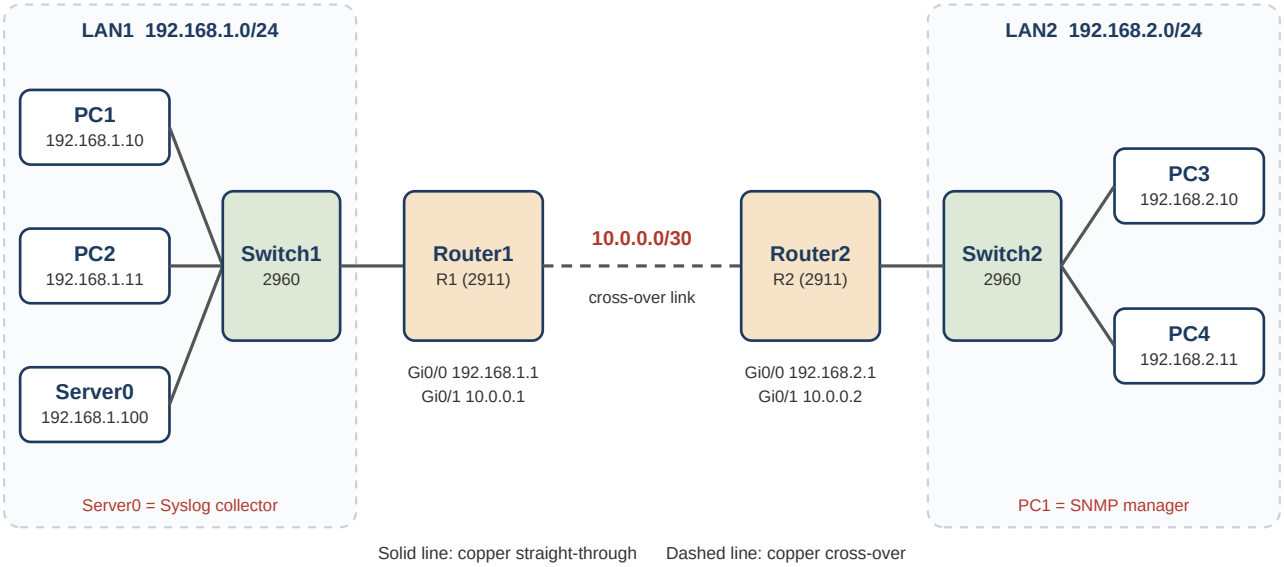


Figure 2: Logical topology with interfaces and IP addressing.

3.2 Device inventory

Device	Model	Hostname	Role
Router1	Cisco 2911	R1	Default gateway for LAN1; RIPv2; SNMP agent; Syslog client
Router2	Cisco 2911	R2	Default gateway for LAN2; RIPv2; SNMP agent; Syslog client
Switch1	Cisco 2960-24TT	-	Layer 2 access switch for LAN1
Switch2	Cisco 2960-24TT	-	Layer 2 access switch for LAN2
PC1, PC2	PC-PT	-	LAN1 clients; PC1 is also the SNMP manager
PC3, PC4	PC-PT	-	LAN2 clients
Server0	Server-PT	-	Syslog collector for both routers

3.3 IP addressing plan

Device	Interface	IP address	Subnet mask	Default gateway
R1	Gi0/0 (to Switch1)	192.168.1.1	255.255.255.0	-
R1	Gi0/1 (to R2)	10.0.0.1	255.255.255.252	-
R2	Gi0/0 (to Switch2)	192.168.2.1	255.255.255.0	-
R2	Gi0/1 (to R1)	10.0.0.2	255.255.255.252	-
PC1	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.1
Server0	FastEthernet0	192.168.1.100	255.255.255.0	192.168.1.1
PC3	FastEthernet0	192.168.2.10	255.255.255.0	192.168.2.1
PC4	FastEthernet0	192.168.2.11	255.255.255.0	192.168.2.1

3.4 Cabling plan

From	To	Cable type
PC1, PC2, Server0	Switch1 (Fa0/2, Fa0/3, Fa0/4)	Copper straight-through
Switch1 Fa0/1	Router1 Gi0/0	Copper straight-through
Router1 Gi0/1	Router2 Gi0/1	Copper cross-over (router to router)
Router2 Gi0/0	Switch2 Fa0/1	Copper straight-through
PC3, PC4	Switch2 (Fa0/2, Fa0/3)	Copper straight-through

Design choices. RIP version 2 was chosen because it exchanges routes automatically, so a failed link or a removed routing process shows up immediately in the routing tables - ideal for the troubleshooting exercises. A /30 subnet is used for the router link because it needs only two usable addresses. Switches run with default settings since both LANs are single-VLAN.

4. Configuration

All devices were configured from the CLI (routers) or the IP Configuration panel (hosts). The complete command listings are in Appendix A; the essential parts are summarised here.

4.1 Interfaces and routing

Router1 (Router2 is configured symmetrically with 192.168.2.1 and 10.0.0.2)

```
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# network 192.168.1.0
R1(config-router)# network 10.0.0.0
R1(config-router)# no auto-summary
```

4.2 Monitoring services

Applied on both routers

```
! Syslog: send router logs to Server0
logging host 192.168.1.100

! SNMP: read-only and read-write community strings
snmp-server community public ro
snmp-server community private rw
```

On Server0 the SYSLOG service was switched on (Services tab). PC1 uses the built-in MIB Browser with address 192.168.1.1 or 192.168.2.1, read community *public* and SNMP version v2c.

5. Monitoring and verification

5.1 Interface and routing tables

The first check after configuration is whether interfaces are up and whether routes are learned. On Router2 both configured interfaces report **up / up**; the third interface and Vlan1 are unused and administratively down by default.

```
Router> enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface gigabitEthernet 0/0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown

R2(config-if)#exit
R2(config)#interface gigabitEthernet 0/1
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown

R2(config-if)#exit
R2(config)#router rip
R2(config-router)# version 2
R2(config-router)# network 192.168.2.0
R2(config-router)# network 10.0.0.0
R2(config-router)# no auto-summary
R2(config-router)#end
R2#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
               ^
% Invalid input detected at '^' marker.

R2#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.2.1     YES manual up                    up
GigabitEthernet0/1     10.0.0.2        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Figure 3: Router2: show ip int brief - Gi0/0 (192.168.2.1) and Gi0/1 (10.0.0.2) are up/up.

On Router1 the routing table contains the two connected networks (C and L entries) and a **R** entry, proving that RIP learned the LAN2 prefix from Router2 across the 10.0.0.0/30 link. The administrative distance and metric are [120/1], i.e. RIP, one hop away.

```
R1(config-if)#exit
R1(config)#router rip
R1(config-router)# version 2
R1(config-router)# network 192.168.1.0
R1(config-router)# network 10.0.0.0
R1(config-router)# no auto-summary
R1(config-router)#end
R1#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
 show ip int brief
              ^
% Invalid input detected at '^' marker.

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:12, GigabitEthernet0/1

R1#
```

Figure 4: Router1: show ip route - 192.168.2.0/24 learned via RIP from 10.0.0.2.

5.2 Ping and traceroute

Ping verifies reachability and round-trip time; traceroute shows the path hop by hop. From PC1 to PC3:

- Ping: replies arrive from 192.168.2.10 with **TTL=126** (initial value 128 minus two router hops). The first request timed out because PC1 had to resolve the gateway MAC address with ARP; the remaining requests succeeded (3/4 received).
- Traceroute: three hops - 192.168.1.1 (Router1), 10.0.0.2 (Router2) and the destination 192.168.2.10 - confirming the path.

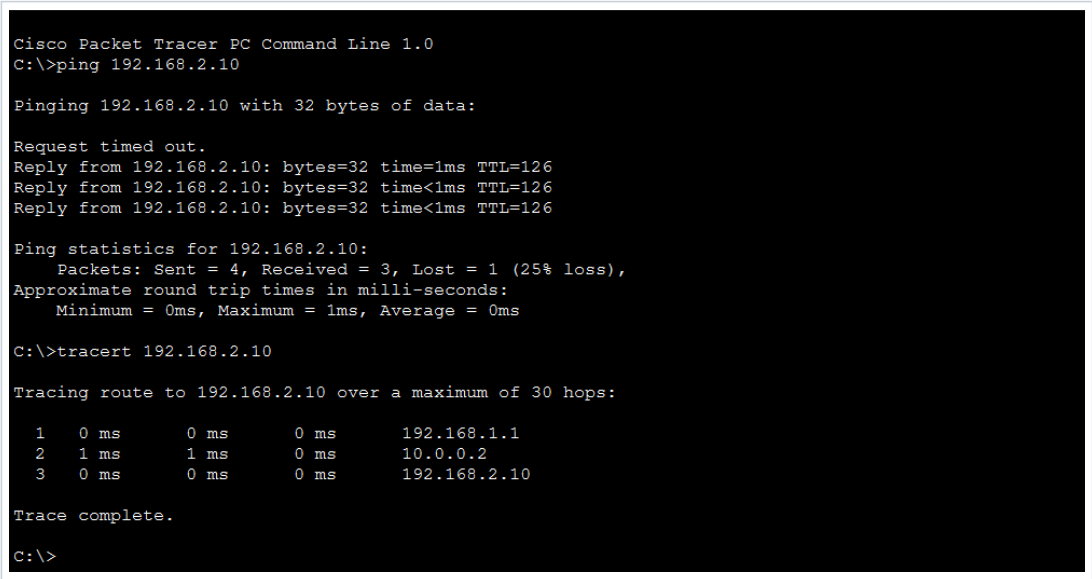


Figure 5: PC1: ping and tracert to PC3 (192.168.2.10).

5.3 Syslog

Both routers send their event messages to Server0. To generate real events, the Gi0/1 interface on Router1 was shut down and brought back up. The Syslog table below shows the configuration change and the link and line-protocol transitions as they were reported, together with the source address of each router.

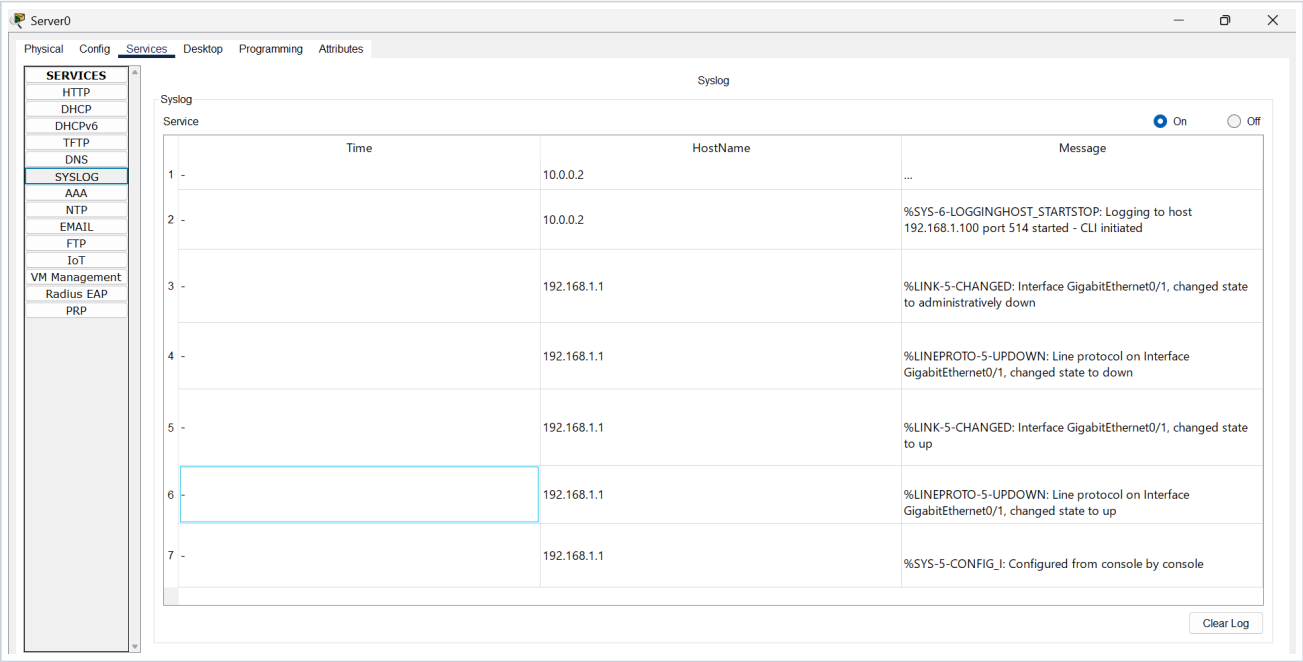


Figure 6: Server0 Syslog service receiving messages from Router1 (192.168.1.1) and Router2 (10.0.0.2).

Message	Meaning
%SYS-6-LOGGINGHOST_STARTSTOP ... started	Router began sending logs to 192.168.1.100 on UDP port 514
%SYS-5-CONFIG_I: Configured from console	Someone changed the configuration from the console
%LINK-5-CHANGED ... administratively down	Interface Gi0/1 was shut down manually
%LINEPROTO-5-UPDOWN ... down	Layer 2 protocol on the interface stopped
%LINK-5-CHANGED ... up / %LINEPROTO ... up	Interface and protocol recovered after no shutdown

Severity levels are encoded in the message code: level 5 is a *notification* and level 6 is *informational*.

5.4 SNMP

SNMPv2c was enabled on both routers. From PC1, the MIB Browser sent **Get** requests to each router. The queried objects belong to the standard MIB-2 system group:

Object	OID	Result
sysDescr.0	.1.3.6.1.2.1.1.1.0	Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M)
sysName.0	.1.3.6.1.2.1.1.5.0	R2 (queried on 192.168.2.1)

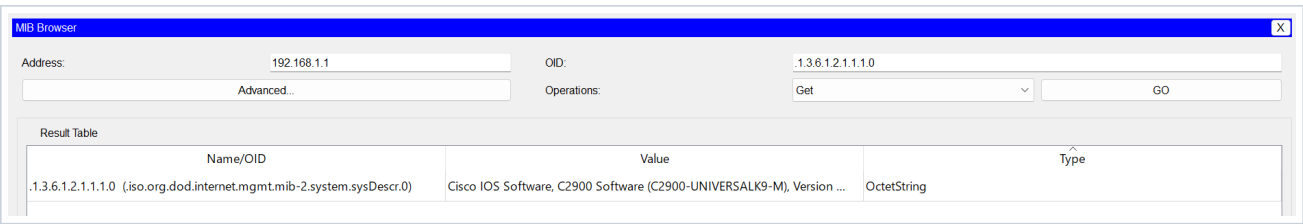


Figure 7: SNMP Get of sysDescr.0 from Router1 (192.168.1.1).

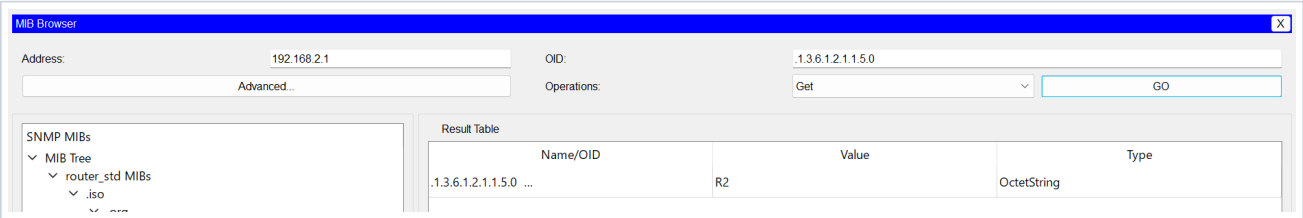


Figure 8: SNMP Get of sysName.0 from Router2 (192.168.2.1).

During setup, the first SNMP request returned *Request timed out* even though ping to the router worked, which means the network path was fine and the problem was in the SNMP settings. After checking the agent configuration on the router and the community string and version in the manager's Advanced settings, the requests succeeded. Typical causes of this symptom are a missing *snmp-server community* line on the router or a community string or version mismatch on the manager.

5.5 Packet-level view (Simulation mode)

Simulation mode, filtered to ICMP, was used to watch a ping from PC1 to PC3 move through the network. The event list records each device the packet visits and the time elapsed. The packet crosses Switch1, Router1, Router2 and Switch2 and the echo reply returns along the same path; the PDU list reports the final status **Successful**. As on the real network, the first attempt may fail while ARP entries are learned.

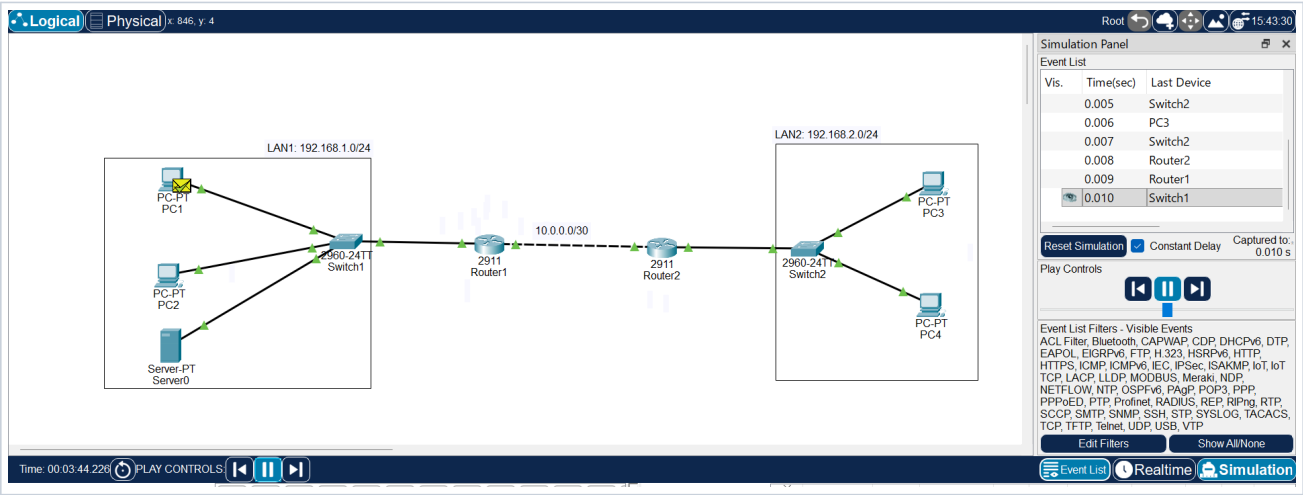


Figure 9: Simulation mode: event list for ICMP between PC1 and PC3, final status Successful.

5.6 Monitoring summary

Tool	What it shows	Where used	Outcome
ping	Reachability, latency, TTL	PC1, PC3	End-to-end connectivity confirmed
tracert	Hop-by-hop path	PC1	Path via 192.168.1.1 and 10.0.0.2
show ip interface brief	Interface status	R1, R2	All used interfaces up/up
show ip route	Learned and connected routes	R1, R2	RIP routes present
Syslog	Device events in one place	Server0	Link and config events collected
SNMP (MIB Browser)	Device identity and status	PC1	Both routers queried remotely
Simulation mode	Packet path and timing	Whole network	Packet flow visualised

6. Troubleshooting

6.1 Method

Each fault was handled with the same repeatable process:

- **Baseline:** record that the network works before changing anything.
- **Identify:** describe exactly what fails and what still works.
- **Isolate:** use the evidence to place the fault on a device or layer (host, switch, router, link).
- **Fix:** apply the smallest change that removes the root cause.
- **Verify:** repeat the original failing test and confirm the network is healthy again.

6.2 Incident 1 - Router interface shut down

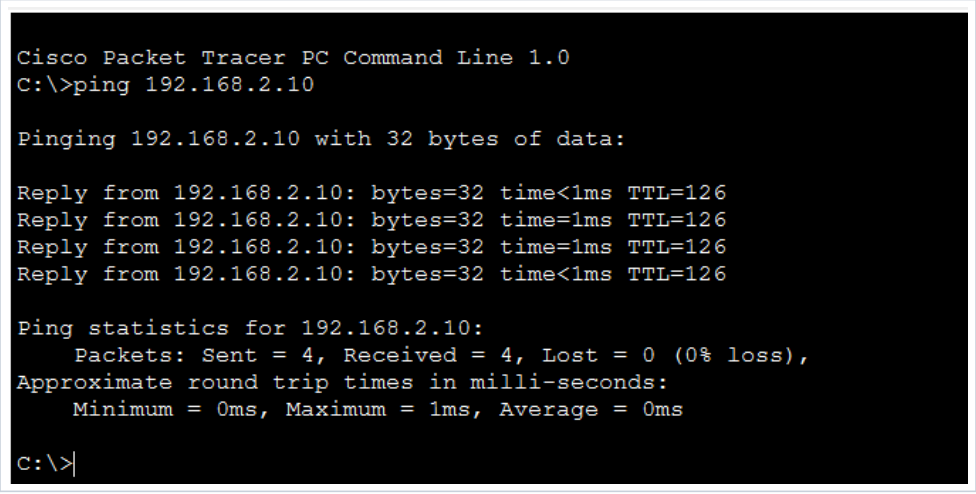


Figure 10: Baseline before the fault: PC1 pings PC3 with 0% loss.

Fault injected on Router1

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# shutdown
```

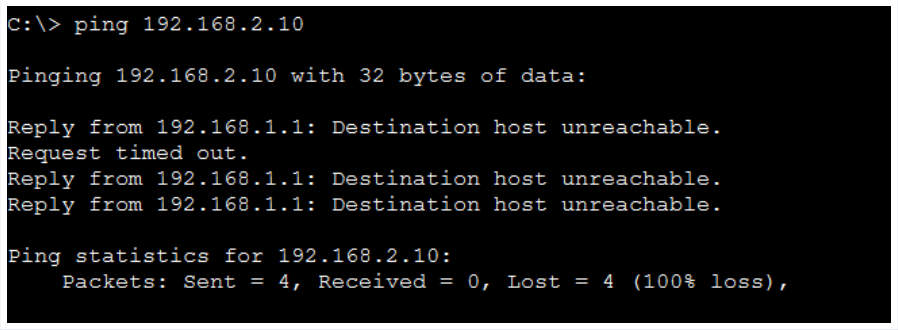


Figure 11: Symptom: ping from PC1 to PC3 fails with 100% loss.

Item	Detail
Symptom	Ping from PC1 to PC3 failed (4 sent, 0 received). The replies were *Destination host unreachable* and came from 192.168.1.1.
Analysis	A reply from 192.168.1.1 means the packet reached Router1 - so PC1, the cable and Switch1 are fine - and Router1 itself reported that it has no route to 192.168.2.0/24. The fault is therefore at Router1 or on the link to Router2, not in LAN1.
Root cause	Interface GigabitEthernet0/1 on Router1 (the link to Router2) was administratively shut down. With the interface down, Router1 lost the RIP route to 192.168.2.0/24.
Resolution	Re-enable the interface with *no shutdown*; RIP then re-learned the route.
Verification	The interface returned to up/up, *show ip route* shows the route to 192.168.2.0/24 again (learned 15 seconds earlier) and ping from PC1 to PC3 succeeds.

Fix

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# no shutdown
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#no shutdown

R1(config-if)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:15, GigabitEthernet0/1
```

Figure 12: After the fix: interface up and the RIP route to 192.168.2.0/24 restored on Router1.

```
C:\> ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 13: Verification: ping from PC1 to PC3 succeeds again (4/4, TTL=126).

6.3 Incident 2 - Wrong default gateway on a host

The default gateway on PC3 was changed from 192.168.2.1 to 192.168.2.254, an address that does not exist.

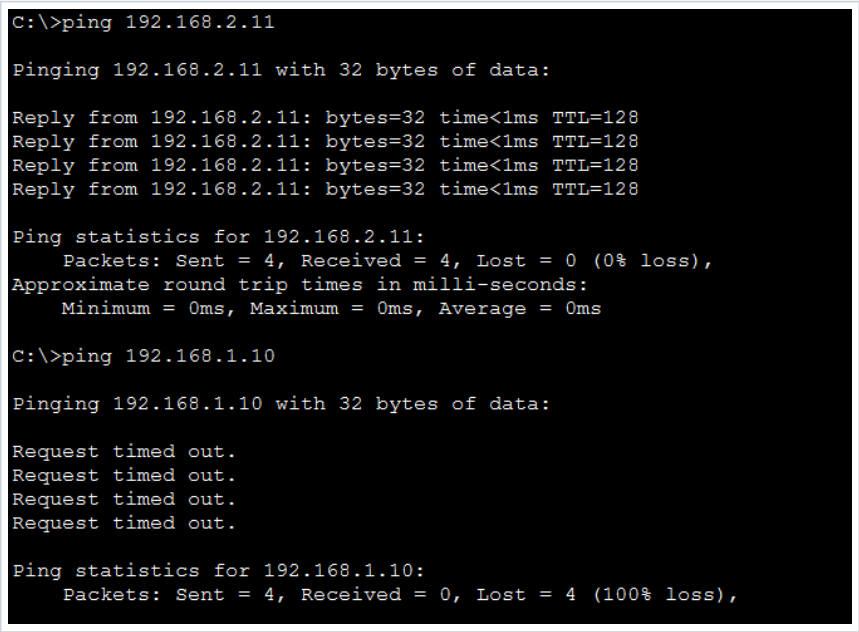


Figure 14: Symptom: PC3 reaches PC4 in its own LAN (TTL=128) but pings to PC1 time out.

Item	Detail
Symptom	PC3 could ping PC4 in the same subnet but could not reach anything outside LAN2 (request timed out).
Analysis	Local traffic works, so the NIC, cable and Switch2 are fine. Remote traffic fails, which points to the default gateway. Unlike Incident 1, no device answered at all: the packets were addressed to a gateway that does not exist.
Diagnosis	Ping to the configured gateway 192.168.2.254 timed out, while ping to the real router interface 192.168.2.1 succeeded (TTL=255, i.e. a router). The router is healthy; only the gateway setting on PC3 is wrong.
Root cause	Incorrect default gateway (192.168.2.254) configured on PC3.
Resolution	Set the default gateway to 192.168.2.1 in IP Configuration.
Verification	*ipconfig* shows 192.168.2.1 as gateway and ping from PC3 to PC1 succeeds (TTL=126).

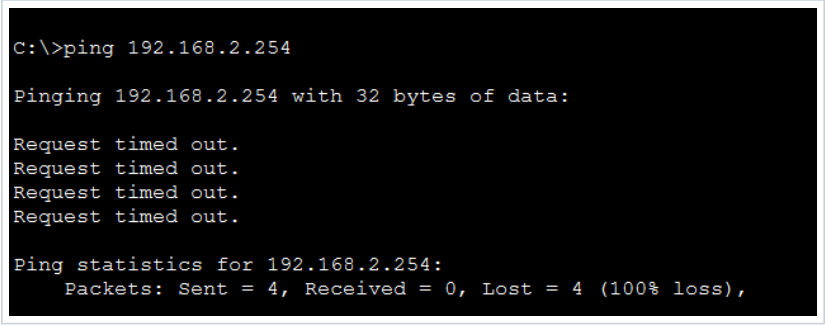


Figure 15: Diagnosis: the wrong gateway 192.168.2.254 does not answer.

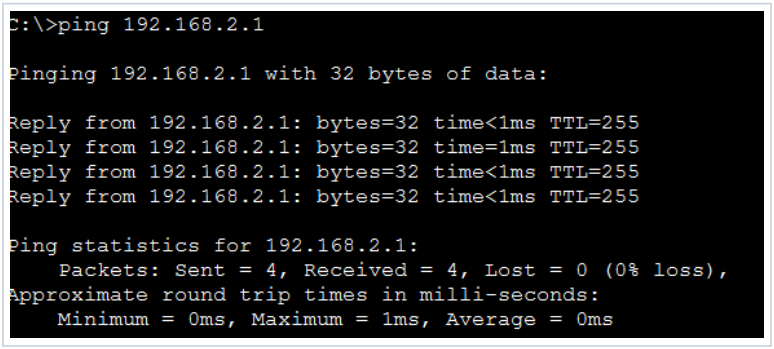


Figure 16: Diagnosis: the real router interface 192.168.2.1 answers (TTL=255).

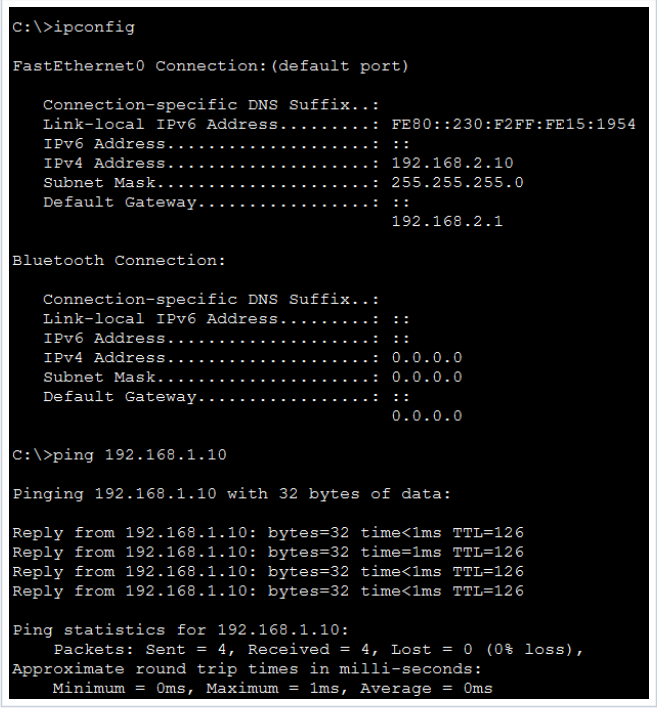


Figure 17: Verification: corrected gateway in ipconfig and successful ping to PC1.

6.4 Incident 3 - Routing process removed from a router

Fault injected on Router2

```
R2# configure terminal
R2(config)# no router rip
```

Item	Detail
Fault	The RIP process was removed from Router2. Router2 stops exchanging routes and discards the routes learned from Router1, so it can no longer forward traffic to 192.168.1.0/24.
Expected indicators	*show ip route* on Router2 shows only connected (C) and local (L) entries and no R entry for 192.168.1.0/24; *show ip protocols* no longer lists RIP.
Root cause	Missing routing configuration on Router2 (dynamic routing disabled), not a physical or addressing problem - all interfaces stayed up.
Resolution	Re-create the RIP configuration on Router2 (commands below) and save it.

Item	Detail
Verification	Router2 again lists 192.168.1.0/24 as a RIP route via 10.0.0.1, learned a few seconds after the configuration was re-applied (see figure), so forwarding between the two LANs is possible again.

Fix

```
R2# configure terminal
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# network 192.168.2.0
R2(config-router)# network 10.0.0.0
R2(config-router)# no auto-summary
R2(config-router)# end
R2# write memory
```

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 192.168.2.0
R2(config-router)#network 10.0.0.0
R2(config-router)#no auto-summary
R2(config-router)#end
R2#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
R    192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:05, GigabitEthernet0/1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0
```

Figure 18: After the fix: Router2 learns 192.168.1.0/24 via RIP from 10.0.0.1.

6.5 Incident summary

#	Fault	Layer / area	Key evidence	Fix
1	Router interface shut down	Network / interface	Unreachable reply from 192.168.1.1; route missing	no shutdown
2	Wrong default gateway on PC3	Network / host configuration	Local ping OK, remote ping times out	Gateway set to 192.168.2.1
3	RIP removed from Router2	Network / routing	No R entries in routing table	Re-create router rip

7. Conclusion and recommendations

The simulation met every objective: a working two-LAN routed network was designed and configured, monitored with ping, traceroute, Syslog, SNMP and Simulation mode, and three different classes of fault were diagnosed and fixed using evidence rather than guesswork. The most useful lessons were:

- The **source of an error message** matters. The unreachable reply from the gateway in Incident 1 immediately narrowed the fault to Router1, while the silence in Incident 2 pointed at the gateway setting.
- Comparing what **still works** (same-subnet ping, router ping) with what fails is the quickest way to locate a fault.
- Centralised logging and SNMP give visibility without logging in to every device.
- A failed first ping is often just ARP; re-test before concluding that something is broken.

Recommended improvements for a production network:

- Replace the default *public*/*private* SNMP community strings, remove the read-write community and prefer SNMPv3.
- Restrict SNMP and Syslog access with access lists and add NTP so log timestamps are meaningful.
- Add a redundant link between the routers so that one failed interface does not split the network.
- Save configurations after every change and keep a backup copy (Appendix A).

Appendix A - Configuration commands

The listings below contain the commands applied to each device. They are the working configuration scripts for this lab and can be pasted into the CLI of an equivalent topology.

Router1 (R1)

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet 0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
exit
router rip
 version 2
 network 192.168.1.0
 network 10.0.0.0
 no auto-summary
exit
logging host 192.168.1.100
snmp-server community public ro
snmp-server community private rw
end
write memory
```

Router2 (R2)

```
enable
configure terminal
hostname R2
interface gigabitEthernet 0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet 0/1
 ip address 10.0.0.2 255.255.255.252
 no shutdown
exit
router rip
 version 2
 network 192.168.2.0
 network 10.0.0.0
 no auto-summary
exit
logging host 192.168.1.100
snmp-server community public ro
snmp-server community private rw
end
write memory
```

End devices (static addressing)

Device	IP address	Mask	Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.11	255.255.255.0	192.168.1.1
Server0 (Syslog ON)	192.168.1.100	255.255.255.0	192.168.1.1
PC3	192.168.2.10	255.255.255.0	192.168.2.1
PC4	192.168.2.11	255.255.255.0	192.168.2.1

Appendix B - Command reference

Command	Purpose
show ip interface brief	Interface status and IP addresses
show ip route	Routing table (C, L, R entries)
show ip protocols	Routing protocols running on the router
show logging	Local log buffer and Syslog settings
ping <ip>	Test reachability and round-trip time
tracert <ip>	Show the path to a destination (PC command prompt)
ipconfig	Show host IP address, mask and gateway
interface <x> / shutdown / no shutdown	Disable or enable an interface
write memory	Save the running configuration